

Open-Source IoT-Integrated SCADA System for Biomass Power Plant Monitoring

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Abstract—This paper presents the development of a condition monitoring system for biomass power plant monitoring, employing the cloud open-source concept combined with notification and security features based on a SCADA system integrated with the IoT. Consequently, the open-source SCADA system is utilized to augment the control system's capabilities. This system will enable interactions among sensors, PLC, SCADA, HMI, and facilitate data storage in a cloud-based database. This integration ensures systematic and organized data collection, fostering efficient data analysis for operational processing within the biomass power plant system. Moreover, it permits rapid problem resolution through notifications via the remote Internet network system.

Keywords—SCADA systems, PLC, IoT, industry 4.0, open source

I. INTRODUCTION

As global energy demand escalates, biomass power plants have emerged as significant contributors, particularly in agricultural nations like Thailand, where raw materials are readily accessible for diverse fuel applications. With the growing pursuit of sustainable energy, the integration of open-source solutions can enhance efficiency, reduce costs, and foster innovation in biomass power generation. Open-source systems offer a collaborative environment that encourages knowledge sharing, flexibility, and adaptability. This paper proposes the development of an IoT-based open-source SCADA system for monitoring biomass power plants, aimed at improving data management and overall operational efficiency of control systems.

SCADA systems represent a form of control in industry, predominantly used for computerized monitoring and management of operational systems. They are capable of tracking and controlling devices in both proximate and distant locations through remote control. Currently, SCADA systems have evolved to communicate with networked computers, enabling online data transmission. This advancement allows for the continuous update of real-time information. SCADA systems operate in conjunction with programmable logic controller (PLC) devices, ensuring seamless bidirectional data exchange [1-2].

The cloud functions as an integrated data management system, consolidating diverse data sources into a unified

storage and infrastructure framework, catering to extensive organizational requirements. Numerous large-scale enterprises contend with complex IT configurations, encompassing various cloud providers, enterprise resources, and legacy software. This complexity often results in duplicate data, inadequate categorization, and strenuous management. The data cloud resolves these issues by eradicating silos and facilitating convenient access for all organizational members to convert data into actionable insights. Pertinent to this paper, data will be gathered from multiple sensors and relayed to the cloud.

In the realm of cloud computing, security is of paramount importance, especially in the context of data storage within public cloud environments. Cloud systems typically employ centralized data storage facilities, rendering them potentially lucrative targets for cyber-attacks. The intricate nature of storage resources, which involve a blend of hardware and software, increases the risk of data exposure in the event of even a minor breach in a public cloud. Consequently, it is prudent to advocate for the utilization of private cloud solutions for the storage of highly sensitive data, wherever feasible. This paper specifically addresses the aspect of information access, underscoring the imperative of robust password protection mechanisms as a fundamental prerequisite for the retrieval of stored data [3].

II. LITERATURE REVIEW

Currently, the communication system (Bus control) connecting PLC, Sensor, HMI, SCADA, and computer systems offers a broad spectrum of options, accommodating the varied expertise and requirements of users. The communication landscape within personal computer networks predominantly relies on Ethernet technology, which has undergone significant evolution into a sophisticated means of data transmission. This technology provides exceptionally high data throughput in the gigabit-per-second range and leverages Application-Specific Integrated Circuits (ASICs) for enhanced processing capabilities in smaller sizes. This development is instrumental in enabling its integration into wireless mobile devices. Moreover, the cost associated with Ethernet hardware is comparatively lower than that of general electronic devices. A significant disparity in performance exists when comparing ethernet and fieldbus communications. Ethernet communication demonstrates superior performance attributes, leading to

a surge in demand for ethernet communications to supersede current fieldbus communication systems. The structural representation of this communication system is depicted in Fig. 1.

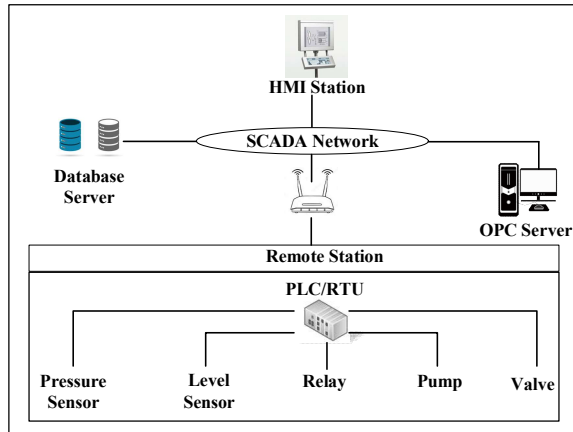


Fig. 1. Network data storage structure using SCADA system.

Fundamentally, ethernet communication cannot be directly applied in industrial automation nondeterministic. This limitation has necessitated the development of new protocols that incorporate deterministic capabilities alongside real-time control functionalities. Industrial ethernet protocols include EtherNet/IP, Profinet, EtherCAT, Modbus, TCP, PowerLink, Sercos III and CC-Link IE.

Industry 4.0 places a significant emphasis on communication, connectivity, and visibility across the entire supply chain [4]. This paradigm shift is expected to yield substantial benefits for a broad spectrum of industries through vertical and horizontal integration across value chains by linking information and data flows from physical and virtual assets [5]. The advent of Industrial and Consumer Internet of Things (IoT) solutions has led to their increasing economic and technological feasibility. In certain instances, these solutions are integral to highly automated production lines. Concurrently, general advancements in science and technology have contributed to a reduction in the costs associated with sensing, processing, and sharing technologies, and as a result, IoT technology has found widespread application across a variety of commercial and industrial products [6]. Smart devices and IoT are instrumental in data generation, while Artificial Intelligence (AI) plays a pivotal role in analyzing and deriving insights from this data to inform decision-making processes. The synergy of these technologies, interconnected to augment value creation, is a cornerstone in the computerization of manufacturing processes to achieve the goal of Industry 4.0

The integration of open-source hardware and software in platforms and devices adheres to a model that enhances

interoperability within the industry, thereby mitigating costly expenses.

As highlighted in the Farnell global IoT survey, improved interoperability has been identified as a key driver in amplifying the benefits of IoT technologies [7]. Consequently, the evolution and implementation of open standards are poised to significantly influence the popularity of open-source platforms. These platforms, rooted in the principles of openness and the utilization of open standards, offer the advantage of being universally accessible and modifiable. This kind of development could potentially catalyze the commercial platforms towards openness, fostering an ecosystem where open-source and commercial platforms can be more flexibly integrated. However, the decision to go towards interoperability involves navigating a complex trade-off such as (1) the profitability of the business case, (2) their strategic position and (3) privacy and security considerations, while fulfilling legal requirements [8].

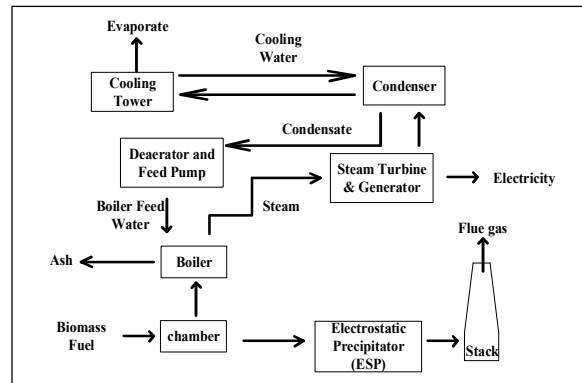


Fig. 2. Electrical energy production process of a biomass power plant

III. THE PROPOSED SCADA SYSTEM FOR BIOMASS POWER PLANT MONITORING

The initial phase of electrical power production in a biomass power plant is shown in Fig. 2 [9]. The process begins with the combustion of biomass fuel to transfer heat to the system's boiler, generating steam. This steam then propels a turbine connected to a generator, thereby producing electricity. The electricity generated is subsequently fed into the Provincial Electricity Authority's grid. A by-product of the combustion process is exhaust gases, which are treated using an electrostatic precipitator before release. Further in the process, the condensate is sent to the condenser by the coolant pump and cools the temperature in the cooling tower for use in the cycle process again. To optimize the plant's efficiency, particularly in terms of reducing drying costs and maximizing electricity production capacity, careful heat integration between the steam power plant and the drying process is essential. For operational optimization, sensor equipment will be installed to collect data on the

machinery's performance. This data will be managed and analyzed via a SCADA system, with additional information collected through a data base cloud. Furthermore, IoT technology will be employed for real-time surveillance and reporting on the status of both sensor devices and machine operations.

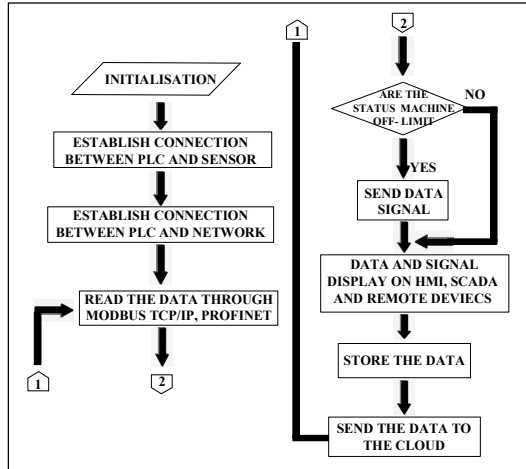


Fig. 3. The flowchart of data communication and remote monitoring using PLC

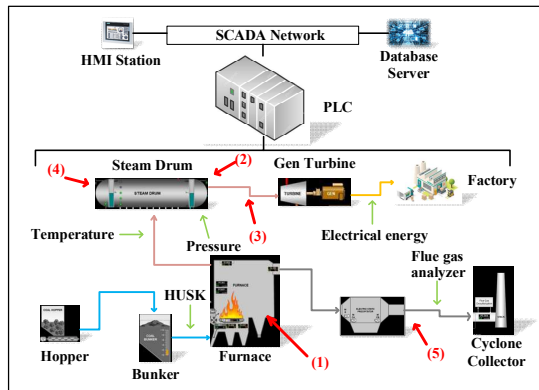


Fig. 4. The preliminary scheme of the process monitoring of biomass power plant (Numbers 1-5 are the locations of installed sensors.)

The data communication and remote monitoring flowchart developed in this research is illustrated in Figure 3. In summary, the designed connection between the programmable logic controller (PLC) is used as that of SIEMES S7-1500 and various sensors PLC connected to the network. It allows data to be received from the machine via the TCP/IP bus and the Profinet interface. Then, if the data is received within a specified time frame, it is displayed on the HMI. Used as HMI Comfort panels or SCADA will It is used as a Rapid SCADA. In addition, this data is sent to cloud storage. It is used as Wecon V-BOX IOT for machine processing and analysis. There are a total of 5 sensors used: (1) temperature sensor, (2) pressure sensor, (3) flow sensor, (4) level sensor, and (5)

exhaust gas air quality sensor. Equipment numbers 1-4 will be of the TURCK brand, while equipment number 5 will be of the DRAEGER brand, as shown in Figure 4. The signal cable is connected to the input side of the PLC to facilitate processing and display. and measurement data management. These cables ensure that measurement data can be efficiently processed, visualized, and managed. Figure 5 shows an example of the installed equipment. Which includes both a water level meter and a flow meter.

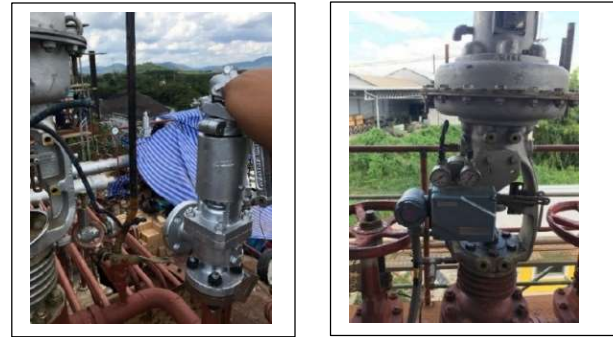


Fig. 5. Actual image of a level sensor (left) and a flow sensor (right)

IV. TEST RESULTS

Functionality of a biomass power plant: The operational process initiates with the preheating of the furnace chamber to a specified temperature. Concurrently, water is introduced into the steam drum chamber to initiate the process, ensuring it attains the predetermined temperature. Subsequent to this preparatory phase, the motor responsible for transporting rice husks (utilized as fuel) is activated, delivering them into the kiln chamber. Following this, the system activates the fan motor to facilitate the mixing of the fuel with air, leading to ignition. Upon the combustion of the rice husks, the temperature within the furnace chamber escalates, inducing the boiling of water in the steam drum, thus generating pressurized steam. Once the specified steam pressure is attained, the pressure valve opens, directing the steam to the generator for electricity generation and utilization of the produced energy.

Turbines in biomass power plants regulate steam pressure to efficiently drive generators. Therefore, it can produce electricity for industrial use. To maintain maximum efficiency It is necessary to control the steam temperature to a specified value before entering the generator. This regulation is achieved by using a water mist spray system. This will effectively cool the steam. The turbine then uses the adjusted steam pressure to control the generator. Steam turbines are single-stage or impulse turbines. The gas turbine is a turboshaft type, converting gas energy from the combustion chamber into rotation and sending mechanical energy out to the shaft. To produce electricity for industrial use during system operation. A comprehensive audit mechanism is in place to detect any anomalies or operational problems. When

such an abnormality is detected, it will be immediately displayed on the confirmation screen. This setting provides important information about various operational situations. This allows for timely intervention and resolution of problems that arise.

In addition, water level sensors and pressure sensors are installed in the steam drum tank to capture and monitor the water level and pressure values. These sensors are needed to collect data inside the tank. Facilitates storage and subsequent analysis. The specific installation locations for these sensors are shown in Figure 6.

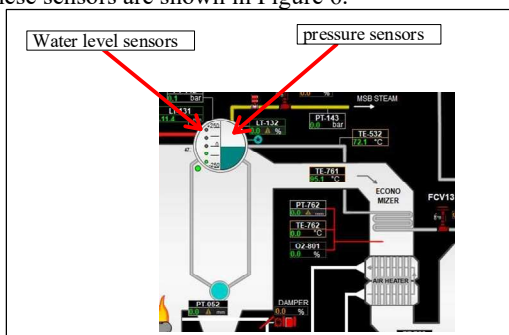


Fig. 6. Water level and pressure sensors are installed in the steam drum tank.

The temperature sensor is strategically positioned inside the furnace to ensure the accurate measurement of temperature values. These measured values play a crucial role in the analysis and monitoring of the biomass combustion rate. Refer to Fig. 7 for the specified installation location.

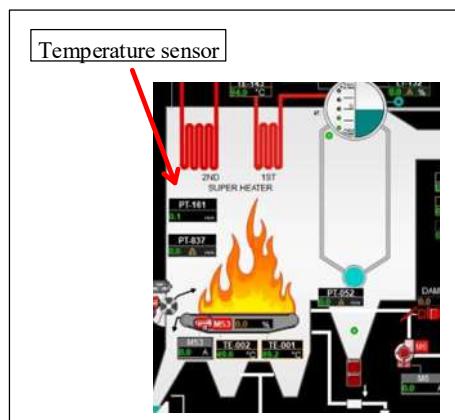


Fig. 7. A temperature sensor is installed inside the furnace.

A flow rate sensor is designated for installation around the water pipe, specifically positioned prior to its entry into the steam drum tank. This water pipe is responsible for transporting water from the deaerator tank. The primary purpose of installing the flow rate sensor is to ensure a consistent water level within the steam drum tank at all times. This consistency is imperative to safeguard the system against potential damage, particularly the risk

of explosion in the steam drum tank, which could result from inadequate water levels. Refer to Fig. 8 for the specific location of the installation.

The carbon dioxide gas measurement sensor is intended to be installed along the pipe connecting the electro station room and the cyclone collector. The primary objective of this sensor is to accurately measure the volume of carbon dioxide gas produced during the combustion process, which originates from the fuel introduced into the furnace. This installation, depicted in Fig. 9, provides a strategic location for accurate measurement of carbon dioxide levels in the system.

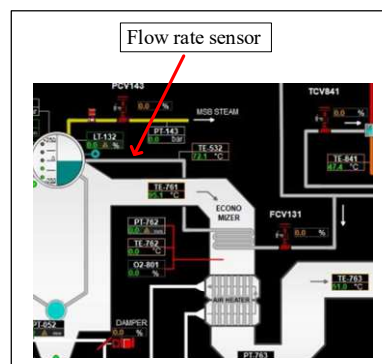


Fig. 8. A water flow sensor is installed inside the steam tank.

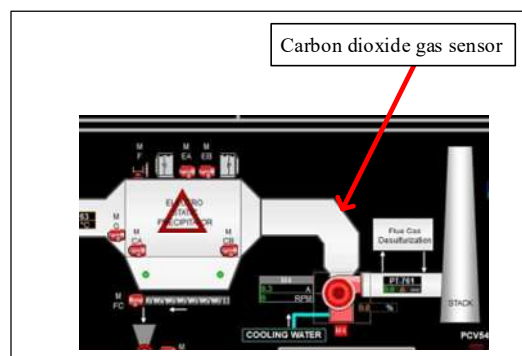


Fig. 9. Carbon dioxide sensors are installed at the power station.

PLC controllers serve as integral components of SCADA hardware due to their compatibility with various digital and analog devices [10]. These controllers play a pivotal role in the acquisition and processing of data received from various sensor devices. After processing, the PLCs transmit work order information back to the devices managing on-site operations. The signals received from sensor devices are processed by the PLC and then transmitted to the SCADA system to present information to the user. Furthermore, there is a feedback loop where commands from the SCADA system are sent to the PLC,

which, in turn, issues commands to on-site loaders for the execution of tasks assigned by users.

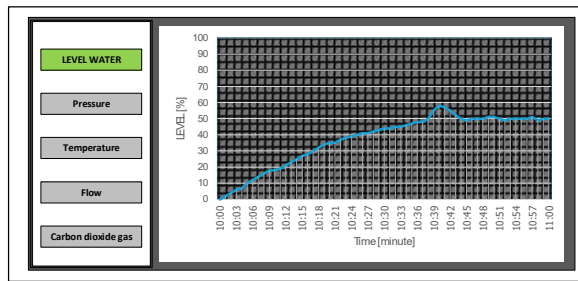


Fig. 10. Example of an analog signal graph received from a water level sensor (displayed on the IoT cloud).

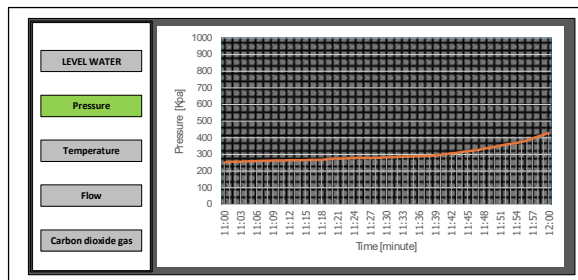


Fig. 11. Example of an analog signal graph received from a pressure sensor (displayed on the IoT cloud).

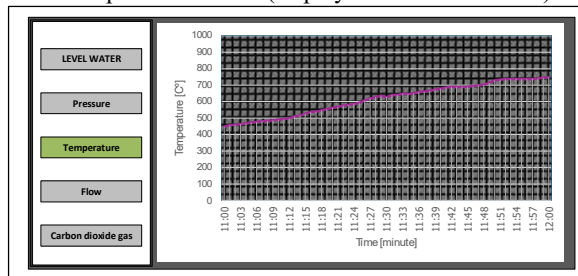


Fig. 12. Example of an analog signal graph received from a temperature sensor (displayed on the IoT cloud).

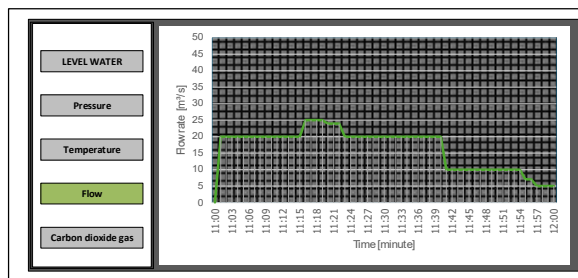


Fig. 13. Example of an analog signal graph received from a flow sensor (displayed on the IoT cloud).

Subsequent to the design and implementation of the SCADA program and display interface, data acquired by the SCADA system is relayed to the PLC for managing load control within the system. The sensor devices transmit data to the PLC, which then stores and updates the information. The system operates in an automatic mode; however, it is also capable of processing manual control decisions to manage the process effectively.

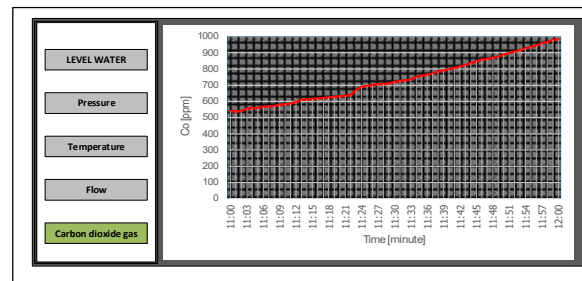


Fig. 14. Example of an analog signal graph received from an exhaust gas air quality sensor (displayed on the IoT cloud).

For optimal combustion efficiency, it is imperative to continuously monitor the fuel level of rice husks to ensure proper mixing with air. The steam pressure process relies on maintaining a consistent water level in the steam drum. Therefore, vigilant monitoring of the steam drum's water level is crucial to avert any potential damage that could arise from insufficient water levels. The control program is designed to analyze and process data from the loading equipment, generating samples that provide essential information on the desired water level. This data, along with other key parameters of the system, are visually represented in various figures: Fig. 10 illustrates water level data, Fig. 11 displays steam pressure information, Fig. 12 shows temperature readings, Fig. 13 details the distance rate, and Fig. 14 presents the concentration of carbon dioxide.

The information collected by the system is subjected to a verification process before being stored in a cloud database for further analysis and monitoring of the system's operation through the IoT surveillance system. Examples of this process are illustrated in Fig. 15 and Fig. 16.

In order to safeguard data integrity and confidentiality, basic encryption techniques are initially used. However, it is important to note that during the process of transmission, the data may assume various formats, necessitating distinct encryption methodologies. Specifically, encryption keys used for data in transit are designed to have a limited lifespan, necessitating periodic updates for enhanced security. Conversely, certain data entries are configured to be retained for longer durations, contributing to the development of a comprehensive and adaptive data protection strategy.

System operation status notifications are sent and notification information is stored in the Cloud IoT system, shown as an example in Fig 17.

A myriad of cryptographic techniques are utilized to encrypt data, substantially bolstering data protection. These techniques play a pivotal role in ensuring the integrity, authentication, and availability of content. In the fundamental process of cryptography, plaintext is transformed into ciphertext using an encryption key. The resulting ciphertext can subsequently be decrypted using a decryption key, as depicted in Fig. 18 [11-12].

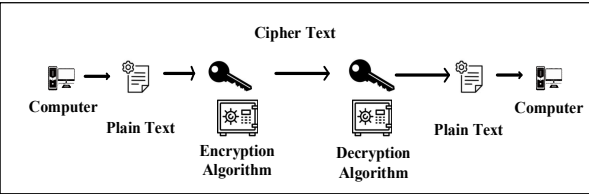


Fig. 18. Basic encryption

V. COUCLUSION

The SCADA system delineated in this paper demonstrates a high degree of precision in monitoring the operational conditions of biomass power plants. It plays a vital role in issuing warnings and maintaining the safety of the plant's operating systems, thereby underscoring its potential and capabilities. Utilizing the cloud open-source concept for systematic data collection, the SCADA system enables efficient processing and analysis of operational data within the biomass power plant infrastructure. The adoption of an open-source framework for the SCADA system not only enhances its functionality but also contributes to significant cost reduction. This cost-effectiveness makes the system particularly advantageous for regional biomass power plants across the country. Furthermore, the versatility of the SCADA system extends its applicability to a range of industries, facilitating equipment monitoring, data collection, and control in diverse environments. These include power generation systems, transmission and distribution networks, and oil and gas production facilities. The SCADA system is instrumental in streamlining data collection, reducing operational costs, time, and complexity. Its remote operation capability minimizes downtime by enabling comprehensive system monitoring and the generation of timely alerts for rapid problem resolution via an internet network. The cloud open-source system is particularly beneficial for SME-level companies, offering them easier access to advanced monitoring technologies. By optimizing work processes in various regions, the SCADA system contributes to overall cost efficiency in operations. Additionally, the implementation of password-protected access ensures robust security of the dataset.

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Real-time Data	Alarm	Historical Data	Cloud SCADA	Lua Script	Pass-through	Configuration
Default group	Group Sec.	Setting				
New Tag	Quick Action	Import/Export	List configuration			
Select All	Status	Name	Value	Port	Read Address	Edit
☐	☒	Water Hight	OFF	Ethernet	M 1.1	Edit Copy Move Delete
☐	☒	Water Low	OFF	Ethernet	M 1.2	Edit Copy Move Delete
☐	☒	Pressure Hight	OFF	Ethernet	M 1.3	Edit Copy Move Delete
☐	☒	Pressure Low	OFF	Ethernet	M 1.4	Edit Copy Move Delete
☐	☒	Temp Hight	OFF	Ethernet	M 1.5	Edit Copy Move Delete
☐	☒	Temp Low	OFF	Ethernet	M 1.6	Edit Copy Move Delete
☐	☒	Alarm Motor 1	OFF	Ethernet	M 1.7	Edit Copy Move Delete
☐	☒	Alarm Motor 2	OFF	Ethernet	M 2.0	Edit Copy Move Delete
☐	☒	Emergency	ON	Ethernet	M 0.0	Edit Copy Move Delete
☐	☒	Status OK	ON	Ethernet	M 0.1	Edit Copy Move Delete

Fig. 15. An example of receiving bit input signals and device operations will be displayed on the cloud IoT system.

Real-time Data	Alarm	Historical Data	Cloud SCADA	Lua Script	Pass-through	Configuration
Default group	Group Sec.	Setting				
New Tag	Quick Action	Import/Export	List configuration			
Select All	Status	Name	Value	Port	Read Address	Edit
☐	☒	Level Water	20.00	Ethernet	MD 8	Edit Copy Move Delete
☐	☒	Temp	500.00	Ethernet	MD 16	Edit Copy Move Delete
☐	☒	Pressure	321.00	Ethernet	MD 24	Edit Copy Move Delete
☐	☒	Flow	21.00	Ethernet	MD 32	Edit Copy Move Delete
☐	☒	Carbon	648.00	Ethernet	MD 40	Edit Copy Move Delete
☐	☒	Motor FD Fan	ON	Ethernet	M 50.1	Edit Copy Move Delete
☐	☒	Motor ID Fan	ON	Ethernet	M 50.2	Edit Copy Move Delete
☐	☒	Motor Pump 1	OFF	Ethernet	M 50.3	Edit Copy Move Delete
☐	☒	Motor Pump 2	OFF	Ethernet	M 50.4	Edit Copy Move Delete
☐	☒	Valve 1	ON	Ethernet	M 50.5	Edit Copy Move Delete

Fig. 16. An example of an analog input signal from the five sensors will be displayed on the cloud IoT system.

Real-time Data	Alarm	Historical Data	Cloud SCADA	Lua Script	Pass-through	Configuration
Alarm List						
Status	Time	Message text				
☒	20/01/23 09:48:02	Temp Water Hight				
☒	20/01/23 08:14:18	Motor 1 Alarm				

Fig. 17. An example of an alarm signal will be displayed on the Cloud IoT system.

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